

Sumaiya Zaman

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Profile

Computer Science and Engineering undergraduate at Chittagong University of Engineering and Technology (CUET) with research experience in Natural Language Processing, Multimodal AI, Computer Vision, and Vision-Language Models. Published research in ACL-affiliated workshops and international conferences, with hands-on experience in deep learning, model fine-tuning, experimental analysis, and AI system development. Skilled in Python, web technologies, databases, REST APIs, and software development practices, with an interest in building reliable AI-powered applications and contributing to machine learning research.

Education

BSc in Computer Science and Engineering — Jan 2023 – Present
Chittagong University of Engineering and Technology (CUET), Chittagong
CGPA through 6th Semester: **3.69 / 4.00**

Higher Secondary Certificate (HSC) — 2019 – 2021
Jalalabad Cantonment Public School and College — GPA: **5.00 / 5.00**

Secondary School Certificate (SSC) — 2017 – 2019
Jalalabad Cantonment Public School and College — GPA: **5.00 / 5.00**

Experience & Leadership

Research Intern (AI Research) — Aug 2026 – Present
TechOptions (BanglaLLM) — Dhaka, Bangladesh (Remote)

- Contribute to Bengali-language AI research, including Bangla NLP/LLMs, data preparation, model evaluation, and language-model experiments.

Organizing Secretary (ML Wing) — Aug 2026 – Present
IEEE Computer Society CUET Student Branch Chapter — Chattogram, Bangladesh

Vice President — Aug 2026 – Present
Sylhet Students' Forum, CUET

Publication Secretary — Aug 2026 – Present
CUET Computer Club

Industrial Attachment Trainee — Sep 2026
Mysoftheaven (BD) LTD. — Dhaka, Bangladesh (On-site)
Completed a three-week industrial attachment.

Research Publications

Published Workshop Papers (2026)

- *CUET_SYNTHETICA@DravidianLangTech 2026: Multi Architecture Transformer Ensemble for Detecting Abusive Tamil Text Targeting Women.* Proceedings of the Sixth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages, ACL 2026.
- *CUET_SYNTHETICA@DravidianLangTech 2026: Multilingual Transformer Based Hope Speech Detection for Coarse and Fine-Grained Classification in Tulu.* Proceedings of the Sixth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages, ACL 2026.
- *CUET_SYNTHETICA@EEUCA 2026: Gated Cross-Modal Attention with Domain-Adapted Text Encoding for Vaccine-Critical Meme Detection.* Proceedings of the 9th Workshop on Event Extraction and Understanding: Challenges and Applications (EEUCA), ACL 2026.

- *Nile Nexus at ImageEval 2026 Shared Tasks: Single-Choice Reformulation and Confidence-Calibrated Refinement for Culturally Grounded Hallucination Detection.* Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks (ArabicNLP 2026). ACL Anthology.
- *Nile Nexus at ArGuard Shared Tasks: Leakage-Controlled MARBERTv2–SVM Ensemble for Arabic Harmful-Prompt Classification.* Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks (ArabicNLP 2026). ACL Anthology.
- *Nile Nexus at BAREC Shared Task 2026: Enhancing Fine-Grained Arabic Readability Assessment through Ordinal-Aware Transformer Ensembling and Decision Calibration.* Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks (ArabicNLP 2026). ACL Anthology.

Conference Papers

- *A Multiclass Polycystic Ovary Syndrome Phenotype Prediction Scheme Using Clinical Data, SHAP, and LightGBM Technique.* RAEEUCCI 2026. DOI: 10.1109/RAEEUCCI67649.2026.11504587
- *Gambit: A Neurological and Neurodevelopmental Disorder Assistant Mobile Application with Symptom Detect and Neurologist Search Features.* IITCEE 2026. DOI: 10.1109/IITCEE67948.2026.11394534
- *A Deep Learning and XAI Based Alopecia Disorder Prediction Scheme Using Temperature Weighted Softmax Ensemble Architectures.* Conference Paper, 2026.
- *An Explainable Machine Learning and Ensemble Stacking Approach Based Multiclass Intellectual Property (IP) Rights Violation Prediction Scheme.* Conference Paper, 2026.

Shared-Task Achievements

- **1st Place** — Hope Speech Detection in Code-Mixed Tulu, Task 1: Coarse-Grained Hope Tone Classification, DravidianLangTech@ACL 2026.
- **1st Place** — Hope Speech Detection in Code-Mixed Tulu, Task 2: Fine-Grained Hope Type Classification, DravidianLangTech@ACL 2026.
- **3rd Place** — Multimodal Identification of Vaccine Content Stance on Social Media, EEUCA 2026; macro-F1: **0.8357**.
- **3rd Place** — BAREC Shared Task, Strict Track, ArabicNLP 2026; QWK: **0.848**.
- **4th Place** — Abusive Tamil Text Targeting Women on Social Media, DravidianLangTech@ACL 2026; macro-F1: **0.8086**.
- **5th Place** — Cross-Cultural Misogynistic Meme Detection (CC-MMD), ICMI 2026 Grand Challenge.
- **5th Place** — ArGuard Shared Task, Arabic Harmful-Prompt Classification, ArabicNLP 2026; macro-F1: **0.860**.
- **6th Place** — ImageEval 2026 AynVQA Shared Task, English Track, ArabicNLP 2026; accuracy: **0.942**.

Technical Skills

Programming: Python, C, C++, JavaScript
Machine Learning / AI: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM, SHAP, Prompt Engineering
NLP / Multimodal AI: Transformer Fine-Tuning, LLM APIs, OCR Text Fusion, Vision-Language Models, CLIP, Multimodal Classification
Web Development: Next.js, React.js, Node.js, Express.js
Database: PostgreSQL, MySQL
API / Testing: REST APIs, Swagger (OpenAPI), Postman
Tools: Git, GitHub, Jenkins, Linux CLI, VS Code
LLM APIs: Google Gemini, Groq, OpenRouter

Selected Software Projects

Virtual Thrift Store *Next.js, Node.js, PostgreSQL, Swagger, Jenkins*
GitHub Repository

- Developed a thrift marketplace with product browsing, backend workflows, and buyer-facing features.
- Designed and documented **39 REST API routes** using Swagger.
- Integrated Gemini, Groq, and OpenRouter APIs for LLM-based product recommendations.

CodeShare Hub *Next.js, Node.js, PostgreSQL, Swagger*
GitHub Repository

- Built a platform for publishing, retrieving, and managing code snippets.

- Added LLM-powered code explanation and main-function extraction.

Mentor–Mentee Scheduling System *React, Node.js, PostgreSQL, Postman*
GitHub Repository

- Implemented authentication, scheduling, appointment booking, feedback, and conflict-handling workflows.
- Tested REST endpoints and booking-conflict cases using Postman.

Additional Projects: Printing Management System (Shell Scripting, Linux CLI); Online Cafe Management System (Web Application).

Awards, Activities & Training

- **Top 20, Solvio AI Hackathon (Sheba Platform):** Developed a credit-risk prediction prototype using XGBoost and ROC-AUC evaluation.
- **Talentpool Grade – Board Scholarship** in both SSC and HSC examinations.
- Participant, CUET CSE Fest 2025 Datathon and IEEE CUET CS Chapter Hackathon.
- Member, CUET Computer Club, IEEE CUET Student Branch, IEEE CUET CS Chapter, and Sylhet Student Forum.
- **Training:** Kaggle Learn – Intro to Machine Learning and Python; DataCamp – Prompt Engineering and Introduction to Python; Udemy – Master Git and GitHub.

Research Interests

Computer Vision, Vision-Language Models (VLMs), Multimodal AI, Multimodal NLP, multilingual NLP, trustworthy AI, misinformation and abusive-language detection, healthcare AI, and LLM-powered systems.

Languages: Bengali (Native) | English (Professional Working Proficiency)